

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We separate the physical generalized-baker section from its symbolic natural extension and realize the physical map by an exact mapping-torus Hamiltonian. Response is constructed from the two noninvertible one-sided Ruelle operators, with bounded Hadamard cut-current insertions for transported observables; no spectral gap is asserted for the bilateral Koopman shift. The full-port work identity and the mechanical valuation coefficient remain explicitly conditional on equality of likelihood cocycles.

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1. ROUND-TEN RECONSTRUCTION: A GENUINE MAPPING-TORUS IMPACT SYSTEM AND A ONE-SIDED RESPONSE CALCULUS

Let $\mathcal{A} = \{1, 2, 3, 4\}$, let $p_i(a) > 0$, $\sum_i p_i(a) = 1$, and assume the branch weights are C^{m+2} on the compact parameter interval under consideration.

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Put $s_i(a) = \sum_{j \leq i} p_j(a)$ and $I_i(a) = [s_{i-1}(a), s_i(a)]$. On the regular square define

$$B_a(q, p) = \left(\frac{q - s_{i-1}(a)}{p_i(a)}, s_{i-1}(a) + p_i(a)p \right), \quad q \in I_i(a).$$

The seam set and all of its iterates have Lebesgue measure zero. We never identify the symbolic natural extension with a symplectic section; it is only a coding of the ordinary two-dimensional section.

1.1. The physical section and its two one-sided factors. Let $\Sigma^+ = \mathcal{A}^{\mathbb{N}_0}$ and $\Sigma^- = \mathcal{A}^{\mathbb{N}}$. For $\omega^+ \in \Sigma^+$, $\omega^- \in \Sigma^-$, define

$$q_a(\omega^+) = s_{\omega_0^+ - 1}(a) + p_{\omega_0^+}(a)s_{\omega_1^+ - 1}(a) + p_{\omega_1^+}(a)p_{\omega_2^+ - 1}(a) + \cdots$$

and the analogous reverse expansion $p_a(\omega^-)$. Let ν_a^+ and ν_a^- be the Bernoulli laws with weights $p(a)$.

Theorem 1.1 (Physical–symbolic factor without a Cantor section). *Outside the countable union of iterated seams, the map*

$$\mathbf{c}_a : (q, p) \mapsto (\omega^+(q), \omega^-(p))$$

is a measurable bijection between the regular physical square and a full-measure subset of $\Sigma^+ \times \Sigma^-$. It satisfies

$$(\mathbf{c}_a)_\#(dq dp) = \nu_a^+ \otimes \nu_a^-, \quad \mathbf{c}_a \circ B_a = (\sigma_+ \times \sigma_-^{-1}) \circ \mathbf{c}_a.$$

In particular Lebesgue measure is invariant for every a , whereas the Bernoulli law varies because the coding map varies.

Proof. For a future cylinder $[i_0 \dots i_r]$, the inverse image under q_a is the half-open interval of length $p_{i_0} \cdots p_{i_r}$; the reverse construction gives the same formula for past cylinders. Product cylinders therefore have Lebesgue measure equal to their Bernoulli product mass. Nested cylinder intervals shrink because $\max_i p_i < 1$ on the compact parameter set. Ambiguity occurs only at endpoints, hence on the iterated seam set. Substitution in the affine branch formula proves the conjugacy. The Jacobian of every branch is one, so the factor identity also proves invariance. $\square$

1.2. An exact autonomous mapping-torus realization. Refine each source strip by the four one-step target strips

$$C_{ij}(a) = \{(q, p) : q \in I_i(a), Q_i(q) \in I_j(a)\}.$$

For every nonempty C_{ij} , take a copy of $[0, 1]_\tau \times C_{ij} \times \mathbb{R}_E$, with symplectic form

$$\Omega = dq \wedge dp + d\tau \wedge dE.$$

Identify its outgoing face $\tau = 1$ with the incoming face of the unique copy containing $B_a(q, p)$, using the exact symplectic map B_a and the identity in E . The quotient is denoted $\mathcal{M}_a$. It is a smooth symplectic manifold on every open stratum and a finite branched symplectic manifold with corners along the physical seam strata.

Theorem 1.2 (Exact autonomous Hamiltonian suspension). *The autonomous Hamiltonian*

$$H_a(q, p, \tau, E) = E$$

on $(\mathcal{M}_a, \Omega)$ has unit return time to the physical incoming section $\{\tau = 0\}$. Its first-return map is exactly B_a on the regular square. Liouville measure induces $dq dp$ on the section. No branch-dependent reset is applied after the Hamiltonian evolution: the branch map is part of the symplectic mapping-torus gluing.

Proof. Hamilton's equations are $\dot{\tau} = 1$ and $\dot{q} = \dot{p} = \dot{E} = 0$ inside every cylinder. At $\tau = 1$, the quotient identification sends the point to the incoming representative $(B_a(q, p), 0, E)$. Since the gluing map preserves $dq \wedge dp$, the forms agree on overlaps and define Ω on the quotient strata. Thus the orbit is a genuine Hamiltonian orbit on $\mathcal{M}_a$, not a flow followed by an external reset. The return and Liouville assertions are immediate. Seam strata have zero Liouville measure, and their oriented branched completion gives the unique one-sided continuation required for boundary tests. $\square$

1.3. Two one-sided spectral gaps. For $0 < \theta < 1$, let $\mathcal{H}_\theta^+$ be the Banach space of bounded future functions with norm

$$\|F\|_\theta = \|F\|_\infty + \sup_{n \geq 0} \theta^{-n} \text{var}_n(F),$$

and define $\mathcal{H}_\theta^-$ analogously on past sequences. Define

$$(P_a^+ F)(x) = \sum_{i=1}^4 p_i(a) F(ix), \quad (P_a^- G)(y) = \sum_{i=1}^4 p_i(a) G(iy).$$

Lemma 1.3 (Correct one-sided Lasota–Yorke estimate). *For either sign,*

$$\text{var}_n(P_a^\pm F) \leq \text{var}_{n+1}(F), \quad \|(P_a^\pm)^k Q_a^\pm F\|_\theta \leq C\theta^k \|F\|_\theta,$$

where $Q_a^\pm F = F - \nu_a^\pm(F)$. The constants and all parameter derivatives through order $m+1$ are uniform on the compact parameter set.

Proof. Prepending one symbol moves every first disagreement one coordinate farther from the origin, which proves the variation estimate. Coupling two copies with the same newly prepended symbols gives contraction by θ^k of the oscillation after k steps. The constant part is removed by Q_a . Parameter derivatives differentiate only the finite weights and therefore obey the same estimate with a finite combinatorial constant. This argument never applies a bilateral inverse shift to a symmetric depth norm, so the translated-cylinder counterexample does not arise. $\square$

Corollary 1.4 (Reduced one-sided resolvents). *The resolvents*

$$S_a^\pm = (I - P_a^\pm)^{-1} Q_a^\pm = \sum_{n \geq 0} (P_a^\pm)^n Q_a^\pm$$

are bounded and C^{m+1} as operators on $\mathcal{H}_\theta^\pm$. Finite products of their derivatives are bounded with the corresponding geometric-series constants.

Proof. This is the norm-convergent geometric series supplied by Lemma 1.3; differentiation is justified by uniform normal convergence. $\square$

1.4. A typed cut-current calculus. For a finite future/past cylinder rectangle $R_w(a)$, define its order-zero functional by integration over the physical rectangle. Its material derivative is defined by the exact one-dimensional Hadamard formula on each moving endpoint. At order k , iterate the formula on the finite stratification generated by the endpoints appearing in the word w . Corner terms are evaluations of traces, never products of unrelated distributions. Let $\mathcal{C}_\theta^r$ be the completion of cylinder functions for the norm

$$\|F\|_{\mathcal{C}_\theta^r} = \sum_w \theta^{|w|} \sum_{j \leq r} \|D_a^j F_w\|_{C^1(R_w)}.$$

Lemma 1.5 (Bounded material current insertions). *For every $k \leq m$, the k -th material derivative of physical cylinder integration extends to a bounded multilinear map*

$$\mathfrak{D}_a^k : \mathcal{C}_\theta^{k+1} \longrightarrow \mathbb{R}.$$

The definition is independent of cylinder refinement, and the sum over all future and past cut depths converges absolutely.

Proof. At fixed word length there are finitely many endpoint and corner strata. Hadamard's formula gives a sum of bulk derivatives and oriented traces, with coefficients made from derivatives of the endpoints $s_i(a)$. If a rectangle is refined, internal faces occur twice with opposite orientations and cancel. The number of strata grows at most exponentially with word length, while the declared $\theta^{|w|}$ coefficient is chosen smaller than the inverse growth constant. Hence the refinement-compatible series is absolutely summable. Since no multiplication of currents is used, all higher-order intersections are ordinary traces on the finite polyhedral stratification. $\square$

Theorem 1.6 (Finite-order physical response for transported observables). *Let $F_a \in \mathcal{C}_\theta^{m+1}$ be a physical observable transported by the coding convention of Theorem 1.1. Then $a \mapsto \int F_a dq dp$ is C^m . Every derivative is the sum of:*

- (i) *the material bulk and cut-current terms of Lemma 1.5; and*
- (ii) *absolutely convergent future and past response words built from $D_a^j P_a^\pm$ and $S_a^\pm$.*

The formula equals the derivative obtained from the common-root quantile coupling of the Bernoulli coordinates.

Proof. Pull the integral to $\Sigma^+ \times \Sigma^-$. Differentiate finite cylinders directly. Weight derivatives give the usual centered score terms; movement of the coding rectangles gives the Hadamard currents. Conditional expectations of the score tails solve the two one-sided Poisson equations, producing the resolvent words. Lemma 1.3 and Lemma 1.5 give a common summable majorant, so cylinder truncation may be removed and derivatives may be iterated. In the common-root coupling, crossing a quantile threshold produces

exactly the same oriented endpoint term, while differentiation away from thresholds produces the bulk term. Hence the two formulas coincide. $\square$

1.5. Full-port work and conditional calibration. On every channel cylinder put the closed work one-form $\eta = c_i d\tau$. Its integral over one return is c_i , including the whole incoming port.

Theorem 1.7 (Mechanical work and conditional coefficient). *The accumulated work over n returns is the symbolic additive current $A_n = \sum_{j < n} c_{\omega_j}$. Its canonical tilt is the Bernoulli member with weights proportional to $p_i(a_0)e^{\vartheta c_i}$. If a cash-additive, time-consistent nonlinear valuation is required to use this same mechanical Radon–Nikodym cocycle, its entropic coefficient equals the mechanical conjugate field. Without that compatibility axiom the valuation coefficient remains external.*

Proof. The work identity follows by integrating $c_i d\tau$ from zero to one. The path likelihood factorizes over returns, giving the tilted branch weights. An entropic valuation has likelihood exponent equal to its coefficient times A_n . Equality with the mechanical likelihood for a nonconstant current forces equality of coefficients. No preference axiom alone is used to infer the mechanical field. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs are used only in the acyclic order recorded in the round-ten dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks; they do not replace independent external mathematical review.

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